

Beyond Audio: Visual Cues to Complement Football Commentary for Fans with Visual Impairments

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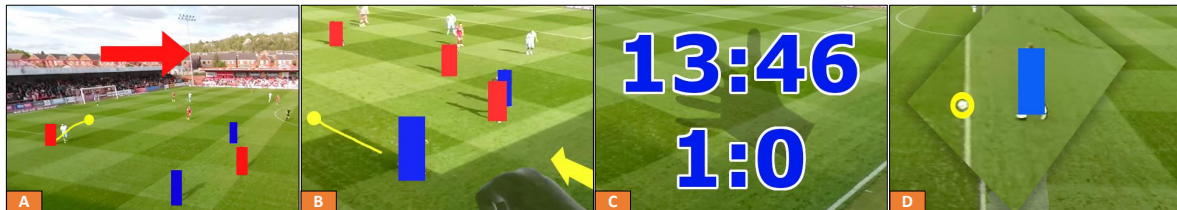


Figure 1: Snapshots illustrating the implemented visual cues for accessible football spectating. (A) A wide view of the pitch with a **counter arrow cue** above indicating the direction of the counter movement. (B) A wrist-attached **directional arrow cue** highlights the positioning of the ball. The ball is emphasized in **full-color yellow** along with its **yellow trajectory trail**. (C) A **hand-attached label** displays high-contrast outlined text of the game time ("13:46") and score ("1:0"), which is invoked by a palm hand gesture. (D) **Magnifying-glass**, activated by finger gun gesture, enlarges part of the pitch showing the **outline of the ball** in yellow. Players are identified by team-specific colored markers.

ABSTRACT

People with visual impairments face unique challenges while attending sports events. Likewise, the stadium atmosphere presents obstacles for using auxiliary interfaces on common media devices. In this work, we present a proof-of-concept AR-based VR application that uses visual cues to highlight key elements of football soccer matches. Following an exploratory approach, we conducted interviews and a qualitative co-design user study with football fans with visual impairments to gather their needs and preferences for such cues. Our findings indicate that AR visual cues can enhance the football spectating experience for fans with visual impairments.

Index Terms: Visual impairment, visual cues, virtual reality, augmented reality, sport events.

1 INTRODUCTION AND RELATED WORK

Football (soccer) is exciting! The fast-paced and unpredictable nature of the sport provides an engaging experience, making it one of the most popular sports worldwide. However, said nature also comes at a cost. Following football matches can be challenging. Even more so in stadium environments, where conventional media tools are difficult to focus on, and factors such as the surrounding atmosphere, weather, and lighting can negatively impact sight. Particularly for people with visual impairments, these challenges bear the risk of making an otherwise enjoyable sport inaccessible. In contrast, AR offers the ability to present contextualized in-situ information. This is important given that most people with visual impairments retain some residual vision, and prior work suggests they often prefer solutions that help them make use of it. [6, 3].

Consequently, researchers have studied different strategies and shown the effectiveness of visual cues for assistive purposes [3, 5].

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However, in the sports context, visual cues in AR have only been used for athletes with visual impairments [4, 1]. The potential of AR to assist spectators with visual impairments remains open. Therefore, we explore accessible football spectating by co-designing a proof-of-concept that overlays simple, customizable cues using hands-free AR technology.

To explore accessible football spectating with residual vision, we first conducted two formative interviews to elicit needs and derive lightweight, customizable visual cues. We then implemented these cues in a VR proof-of-concept and evaluated it in a small user study with three participants with visual impairments (two in-person, one remote). Our findings suggest that simple ball-centric cues are promising, while preferences for individual cues vary across users. Based on our findings, we suggest that future work should manage visual load to complement audio commentary.

2 METHODOLOGY

Our procedure is structured in threefold. Firstly, we conducted interviews with football fans with visual impairments to gain a better understanding of their needs. Second, based on the interview findings, we developed a set of visual cues. Lastly, our prototype was tested in a qualitative co-design user study with three participants. The participants were aged 22–44. P1 (2% visual acuity) and P3 (10–20% visual acuity) are both diagnosed with cone-rod dystrophy. P2 (10–15% visual acuity) is diagnosed with retinitis pigmentosa.

2.1 Interviews

P1 and P2 took part in the interviews. During the interviews, participants were asked about their experiences watching football at home and in stadiums, as well as with AR and VR devices. The sessions were analyzed through iterative inductive coding [2], which captured the central thoughts of the participants regarding visual approaches to assistive tools.

2.2 Visual Cue Design

We implemented a set of features for accessible spectating based on the results of our interviews. Each of the cues is shown in Figure 1. P1 had difficulty tracking the ball, suggesting a borderline high-light, which motivated highlighting the **outline of the ball**. The participants emphasized that overlays should remain readable and

avoid clutter (“*less is more*”), proposing a **hand-attached label** that displays the match time and score when the user looks into their palm. Following this approach, we implemented **player highlighting** to differentiate players of different teams by color in spatially tight situations such as corners. To support fast, hard-to-follow transitions in play, we implemented a **counter arrow cue** during large and fast movements. Finally, P1 pointed out that needs vary strongly with visual acuity, inspiring different modes for vision levels. We included the possibility of a **magnifying-glass**, which is activated using a finger gun hand gesture. Likewise, we included **full-color cues** for ball tracking and supplementary cues such as the **ball trajectory trail and directional arrow**, as well as full customization of color and size by the user.

2.3 User Study

To test the application, we showed the participants 360° videos of football matches in a qualitative user study. P1, P2 and P3 took part in this study. During the study sessions, participants were free to express their opinions on the application and the visual cues it provided. We used Unity 6 and the Meta Quest 3. Participants were seated and used their hands for interaction. All the interviews and study sessions were recorded for their analysis through a thematic analysis approach [2]. The study sessions were inductively analyzed and interpreted to identify key topics and recurring patterns. Through iterative cycles, we refined the key topics to ensure they accurately reflected the participants’ feelings. During the analysis, we continuously categorized the findings to gain a deeper understanding of how individual statements contribute to broader themes.

3 RESULTS

Participants reported that shape-based visual cues supported their ability to follow football plays, particularly in locating and tracking the ball. All participants found highlighting the ball more useful than highlighting the players. Preferences for full color versus outline differed across users. Supplementary cues, such as the trail or transition arrow, were helpful for some participants but not noticeable or even distracting for others. Participants valued access to detailed contextual information, especially game time and score, which they explicitly requested in the interviews and responded positively to in the user study. However, this information was only readable for users with more residual vision, and some relied on magnification to access details. As a result, information retrieval was mixed and depended on visual acuity and scene conditions. Participants stressed that visual cues should not replace auditory cues, but rather complement them. Customization was appreciated by all participants, with emphasis on the importance of adjusting the cues and text to match individual needs. Overall, the cues received positive feedback, and participants highlighted the potential of combining visual cues with existing auditory cues. As P2 stated, “*In conjunction with audio description in the stadium, this is definitely what is still missing for visually impaired people.*”

4 DISCUSSION AND CONCLUSION

Our results suggest that lightweight, shape-based visual cues can assist football fans with visual impairments, particularly by supporting ball tracking. We indicate that highlighting the ball can meaningfully support understanding the game and should be considered an important design choice. Participants noted that the cues are not experienced in isolation: the usefulness often depended on how cues work together in practice. For example, player highlighting, combined with the counter arrow, improved understanding in spatially tight situations for some, whereas others found this combination cluttering or confusing. Overall, further implementations must weigh the trade-offs between visuals and distractions.

While participants repeatedly emphasized the value of detailed contextual information, our findings indicate that visual presenta-

tion alone has limitations. Visual cues augment information; however, the effectiveness diminishes with declining visual acuity, underscoring the importance of audio. Thus, visual cues should never replace auditory cues, but rather complement them to create a rich viewing experience.

These findings highlight the potential of AR-based visual cues in real stadiums to give fans with visual impairments more direct, moment-to-moment access to what is happening on the pitch.

Since this work only offers a proof-of-concept for visual cues in accessible sports spectating, future research should integrate these cues into real-life settings. The next step for real football matches is an AR application that displays cues in real-time in large-scale venues, such as stadiums. However, current AR devices still lack the computational power for such use [7]. Future work should also address interaction and customization, and complement visual cues with auditory channels such as audio-descriptive commentary and other forms of sonification.

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